

Homework #1: Solving Practice Problems for Discrete Random Variables

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Problem 1

1. The random variable V has PMF:

$$P_V(v) = \begin{cases} cv^2, & \text{if } v = 1, 2, 3, 4, \\ 0, & \text{otherwise} \end{cases}$$

(a) Find the value of the constant c

Since $P_V(v)$ is a probability mass function, the total probability must equal 1:

$$\sum_{v=-\infty}^{\infty} P_V(v) = 1.$$

Thus,

$$\sum_{v=1}^4 cv^2 = 1$$

$$\Rightarrow c(1^2 + 2^2 + 3^2 + 4^2) = 1$$

$$\Rightarrow c(1 + 4 + 9 + 16) = 1$$

$$\Rightarrow c30 = 1$$

Therefore,

$$c = \frac{1}{30}$$

Problem 1

1. The random variable V has PMF:

$$P_V(v) = \begin{cases} \frac{1}{30}v^2, & \text{if } v = 1,2,3,4, \\ 0, & \text{otherwise} \end{cases}$$

(b) Find $\Pr [V \in \{u^2 \mid u = 1,2,3,\dots\}]$

The set $\{u^2 \mid u = 1,2,3,\dots\}$ represents the set of perfect squares: $\{1,4,9,\dots\}$.

Since V can only take values $\{1,2,3,4\}$, the values that are perfect squares are **1,4**.

Thus,

$$\begin{aligned} \Pr[V \in \{u^2 \mid u = 1,2,3,\dots\}] &= \Pr[V = 1] + \Pr[V = 4] \\ &= \frac{1}{30}(1^2 + 4^2) \\ &= \frac{17}{30} \end{aligned}$$

$$\Pr(V \in \{u^2 \mid u = 1,2,3,\dots\}) = \frac{17}{30}$$

Problem 1

1. The random variable V has PMF:

$$P_V(v) = \begin{cases} \frac{1}{30}v^2, & \text{if } v = 1,2,3,4, \\ 0, & \text{otherwise} \end{cases}$$

(c) Find the probability that V is even

(d) Find $\Pr[V > 2]$

V is even when $V \in \{2,4\}$. Therefore,

$$\begin{aligned} \Pr[V \in \{2,4\}] &= \Pr[V = 2] + \Pr[V = 4] \\ &= \frac{1}{30}(2^2 + 4^2) \\ &= \frac{20}{30} = \frac{2}{3} \end{aligned}$$

$$\Pr(V \text{ is even}) = \frac{2}{3}$$

The values satisfying $V > 2$ are 3,4. Thus,

$$\begin{aligned} \Pr[V \in \{3,4\}] &= \Pr[V = 3] + \Pr[V = 4] \\ &= \frac{1}{30}(3^2 + 4^2) \\ &= \frac{25}{30} = \frac{5}{6} \end{aligned}$$

$$\Pr(V > 2) = \frac{5}{6}$$

Problem 2

2. In each at-bat in a game, mighty Casey swings at every pitch. The result is either a home run (with probability $q = 0.05$) or a strike. Of course, three strikes and Casey is out.

Let Y_1, Y_2, Y_3 be **i.i.d. Bernoulli random variables** with parameter q , • $Y_t \in \{0,1\}$

$$Y_t = \begin{cases} 1, & \text{w.p. } q, \\ 0, & \text{w.p. } 1 - q \end{cases}$$

$Y_t = 1$ denotes a home run, and $Y_t = 0$ denotes a strike on the $t \in \{1,2,3\}$ -th **potential swing**.

The at-bat ends when: a home run occurs, or three strikes are reached.

Define the stopping time

$$N = \min \{t \in \{1,2,3\} : Y_t = 1 \text{ or } t = 3\}$$

Then the **actual outcome** is the stopped sequence $(Y_1, \dots, Y_N)$.

Thus, the **sample space** is:

$$\Omega = \{(1), (0,1), (0,0,1), (0,0,0)\}$$

Problem 2

2. In each at-bat in a game, mighty Casey swings at every pitch. The result is either a home run (with probability $q = 0.05$) or a strike. Of course, three strikes and Casey is out.

(a) What is the probability $\Pr[H]$ that Casey a home run?

The sample space is: $\Omega = \{(1), (0,1), (0,0,1), (0,0,0)\}$

The event H is given by $H = \{(1), (0,1), (0,0,1)\} = \{(0,0,0)\}^c$

Therefore, $\Pr[H] = 1 - \Pr[\{(0,0,0)\}^c] = 1 - (1 - 0.05)^3 = 0.142625$

$$\Pr[H] = 0.142625$$

Problem 2

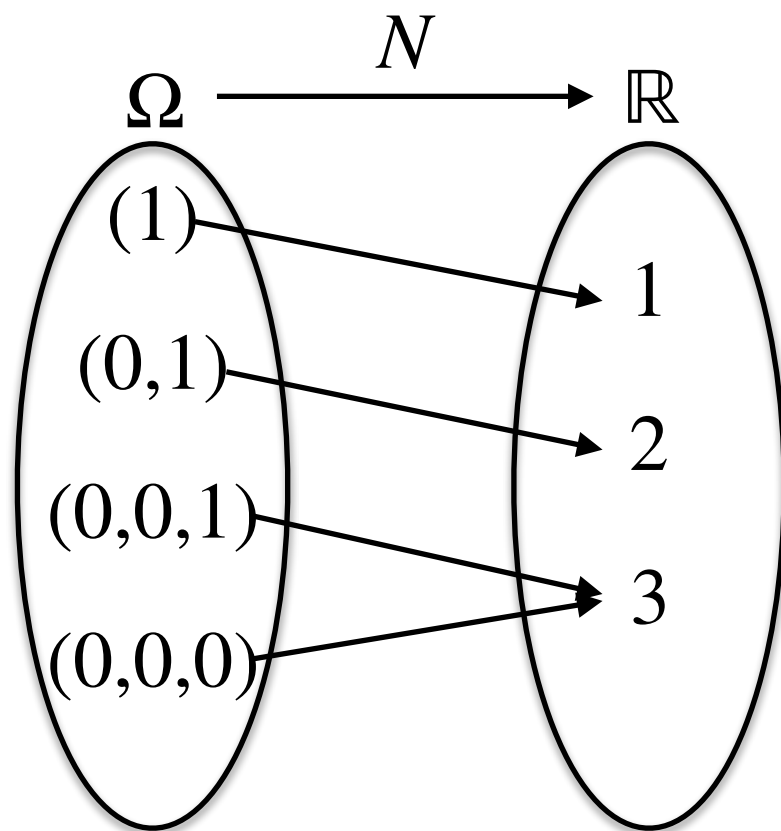
(b) For one at-bat, what is the PMF of N , the number of times Casey swings hit bat?

The **Sample space** is: $\Omega = \{(1), (0,1), (0,0,1), (0,0,0)\}$. The **support of N** is: $\text{supp}(N) = \{1,2,3\}$.

The random variables $N : \Omega \rightarrow \mathbb{R}$ are given by

$$N(\omega) = |\omega|$$

- $|\omega| \triangleq$ The length of sequence ω
- $\text{supp}(N) \triangleq \{n : P_N(n) > 0\}$



Recall that the outcomes formed a **sequence of i.i.d. Bernoulli random variables**. So, the probability of a singleton set is as follows:

$$\Pr[\{(1)\}] = q, \Pr[\{(0,1)\}] = q(1 - q),$$

$$\Pr[\{(0,0,1)\}] = q(1 - q)^2, \Pr[\{(0,0,0)\}] = (1 - q)^3.$$

Therefore,

$$P_N(n) = \begin{cases} q(1 - q)^{n-1}, & \text{if } n = 1, 2, \\ q(1 - q)^2 + (1 - q)^3, & \text{if } n = 3, \\ 0, & \text{otherwise} \end{cases}$$

Problem 2

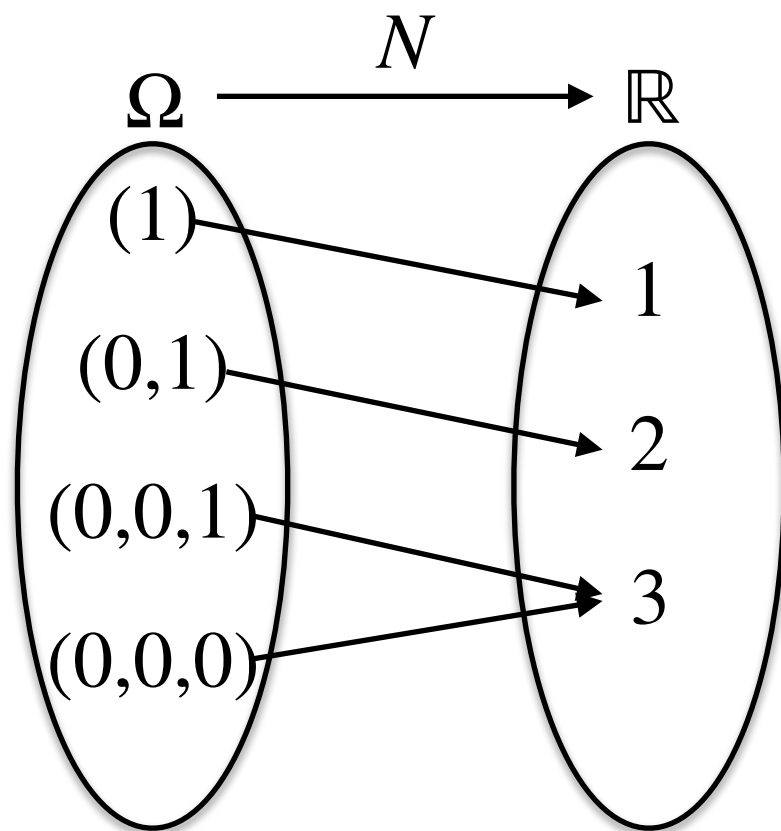
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Therefore,

$$P_N(n) = \begin{cases} 0.05, & \text{if } n = 1, \\ 0.0475, & \text{if } n = 2, \\ 0.9025, & \text{if } n = 3, \\ 0, & \text{otherwise} \end{cases}$$

Problem 3

3. Suppose when baseball player gets a hit, a single is twice as likely as double, which is twice as likely as a triple, which is twice as likely as a home run. Also, the player's batting average, i.e., the probability the player gets a hit, is 0.300. ~

We define the sample space Ω as the set of all possible outcomes in an at-bat, where each outcome is one of the following: a single s , a double d , a triple t , a home run h , or out o . Thus,

$$\Omega = \{s, d, t, h, o\}$$

The probability of singleton set is as follows:

$$\Pr[\{s\}] = 8g, \Pr[\{d\}] = 4g, \Pr[\{t\}] = 2g, \Pr[\{h\}] = g, \Pr[\{o\}] = 0.7$$

where g denotes the probability that the outcome is a home run.

Since the probability that the player gets a hit is 0.3, we have

$$\Pr[\{o\}] = \Pr[\{s, d, t, h\}^c] = 1 - 0.3 = 0.7,$$

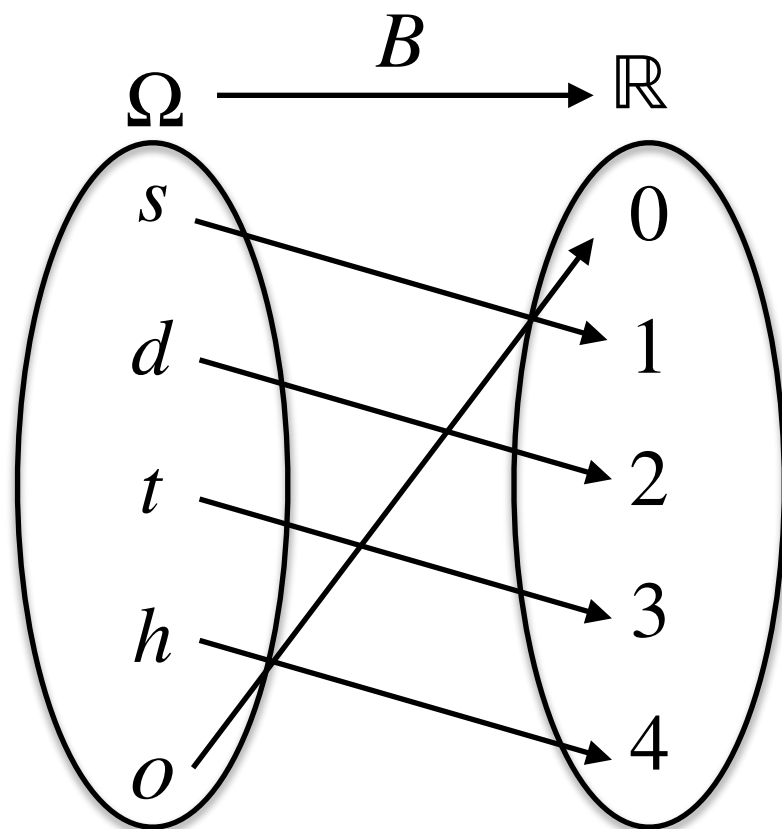
$$\begin{aligned} \Pr[\{s, d, t, h\}] &= \Pr[\{s\}] + \Pr[\{d\}] + \Pr[\{t\}] + \Pr[\{h\}] = 8g + 4g + 2g + g = 15g = 0.3 \\ &\Rightarrow g = 0.02 \end{aligned}$$

Problem 3

3. ~. Let B denote the number of bases touched safely during an at-bat. For example, $B = 0$ when the player makes an out, $B = 1$ on a single, and so on. What is the PMF of B ?

$$\Omega = \{s, d, t, h, o\}$$

$$\Pr[\{s\}] = 0.16, \Pr[\{d\}] = 0.08, \Pr[\{t\}] = 0.04, \Pr[\{h\}] = 0.02, \Pr[\{o\}] = 0.7$$



The random variables $B : \Omega \rightarrow \mathbb{R}$ are given by

$B(\omega)$ = The number of bases touched safely during an at-bat

Therefore,

$$P_B(b) = \begin{cases} 0.7, & \text{if } b = 0, \\ 0.16, & \text{if } b = 1, \\ 0.08, & \text{if } b = 2, \\ 0.04, & \text{if } b = 3, \\ 0.02, & \text{if } b = 4, \\ 0, & \text{otherwise} \end{cases}$$

Problem 4

4. A packet received by your smart phone is error-free with probability **0.95**, independent of any other packet. In one hour, your smart phone receives **12,000** packets. Let X equal the number of packets received with errors. What is $\mathbb{E}[X]$?

$$P_X(x) = \begin{cases} \binom{N}{x} p^x (1-p)^{N-x}, & \text{if } x = 0, \dots, N, \\ 0, & \text{otherwise} \end{cases} : \text{binomial random variable } (N = 12000, p = 0.05)$$

$$\mathbb{E}[X] \triangleq \sum_{x \in \mathbb{Z}} x P_X(x) = \sum_{x=0}^N x \frac{N!}{(N-x)!x!} p^x (1-p)^{N-x} = \sum_{x=1}^N \frac{N!}{(N-x)!(x-1)!} p^x (1-p)^{N-x}$$

Let $k \triangleq x - 1$

$$= \sum_{k=0}^{N-1} \frac{N!}{(N-1-k)!(k)!} p^{k+1} (1-p)^{N-1-k} = Np \sum_{k=0}^{N-1} \frac{(N-1)!}{(N-1-k)!(k)!} p^k (1-p)^{N-1-k}$$

$$= (p + (1-p))^{N-1} = 1$$

$$= Np = 12000 \cdot 0.05 = 600$$

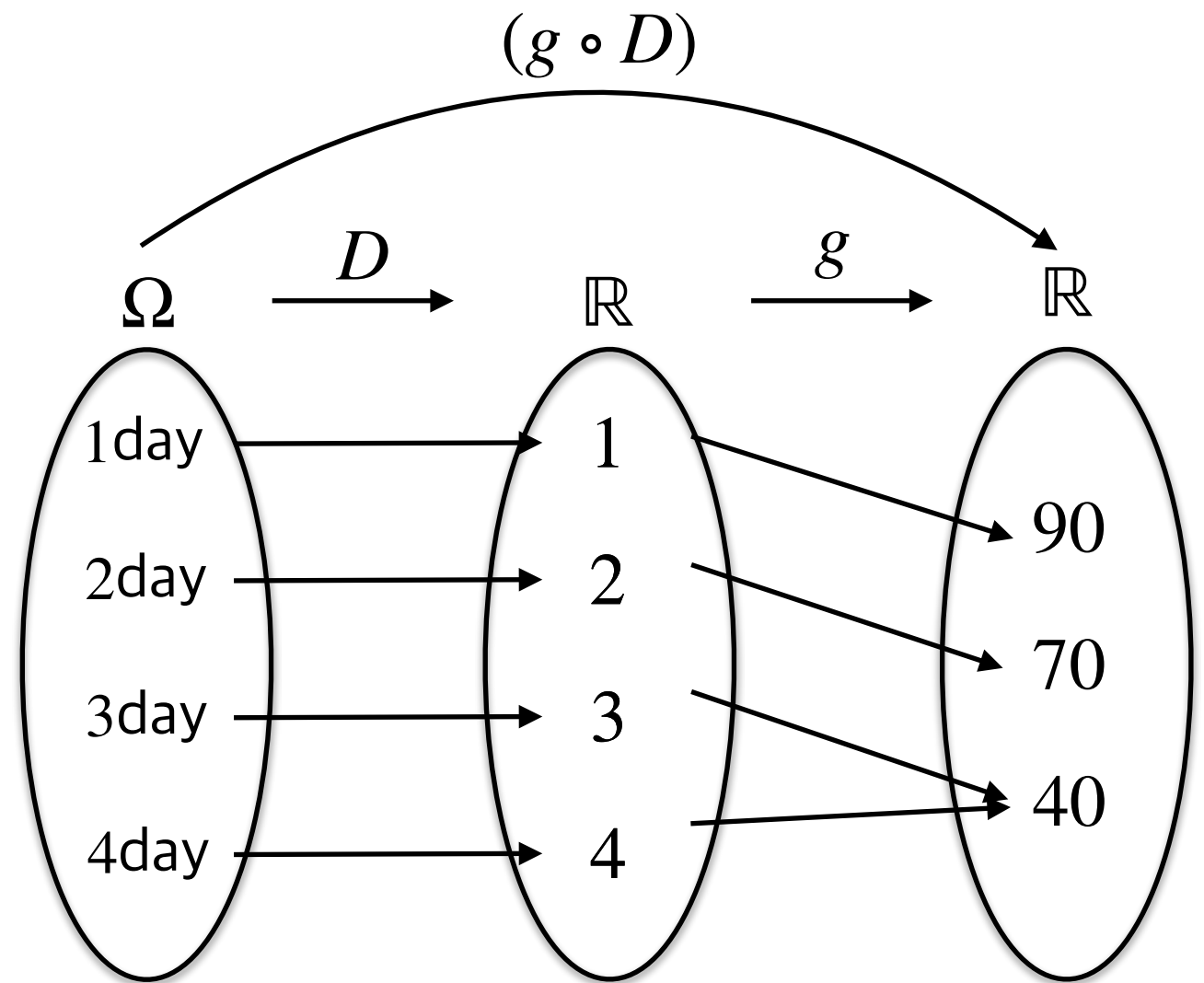
$\mathbb{E}[X] = 600$

Problem 5

5. It can take up to four days after you call for service to get your computer repaired. The computer company charges for repairs according to how long you have to wait. The number of days D until the service technician arrives and the service charge C , in dollars, are described by

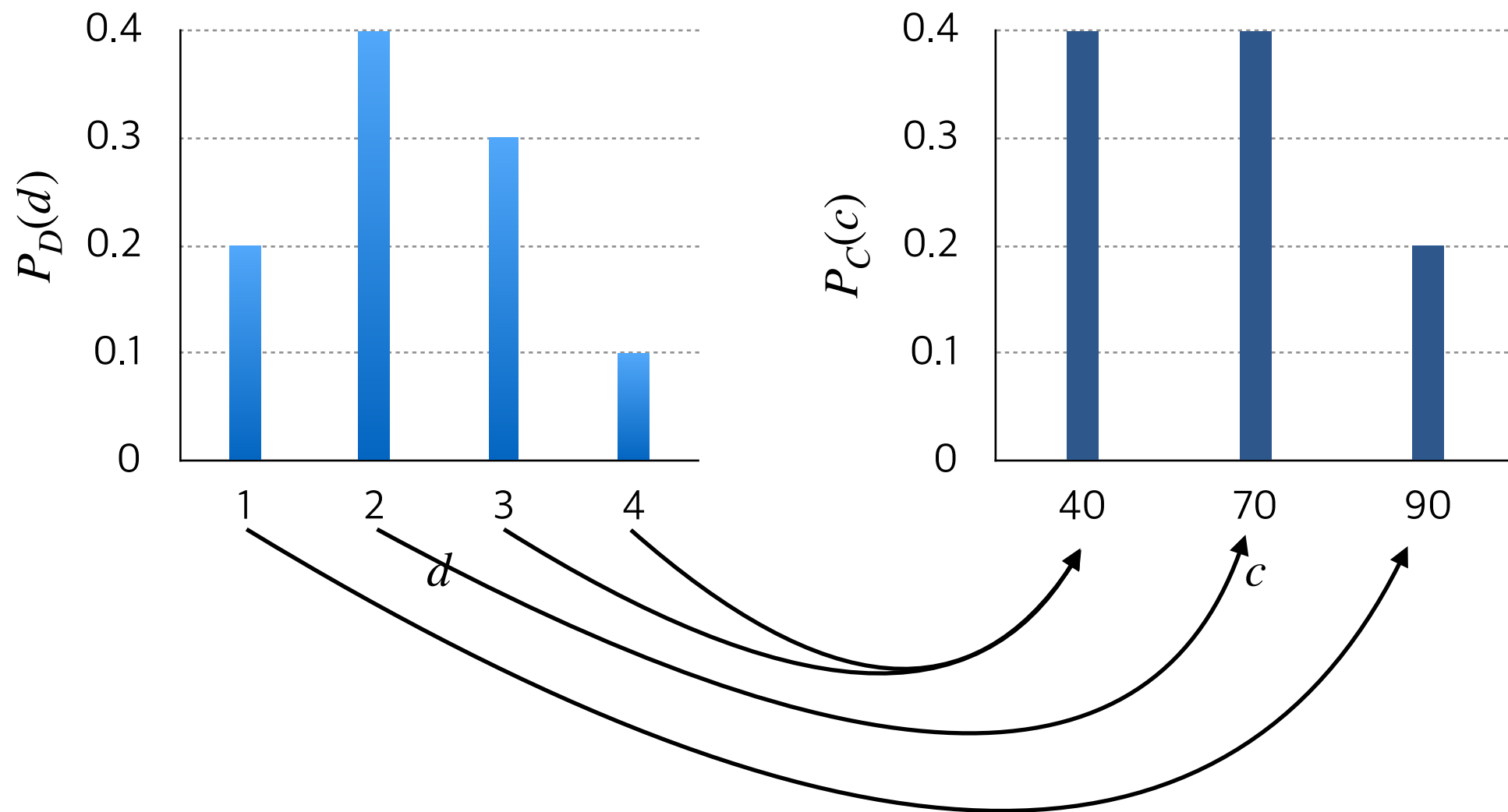
(c) Express C as a function of D

$$C : \Omega \xrightarrow{D} \mathbb{R} \xrightarrow{g} \mathbb{R}$$



Problem 5

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Problem 5

5.

$P_D(d)$

d	1	2	3	4
PMF	0.2	0.4	0.3	0.1

$P_C(c)$

c	90	70	40
PMF	0.2	0.4	0.4

(a) What is the expected waiting time $\mu_D = \mathbb{E}[D]$

$$\mu_d = \sum_{d=1}^4 dP_D(d) = 1 \cdot 0.2 + 2 \cdot 0.4 + 3 \cdot 0.3 + 4 \cdot 0.1 = 2.3$$

$$\mu_D = 2.3$$

(b) What is the expected deviation $\mathbb{E}[D - \mu_D]$

$$\mathbb{E}[D - \mu_D] = \mathbb{E}[D] - \mathbb{E}[\mu_D] = \mu_D - \mu_D = 0$$

$$\mathbb{E}[D - \mu_D] = 0$$

Problem 5

5.

$P_D(d)$

d	1	2	3	4
PMF	0.2	0.4	0.3	0.1

$P_C(c)$

c	90	70	40
PMF	0.2	0.4	0.4

(d) What is the expected waiting time $\mathbb{E}[C]$

$$\mathbb{E}[C] = \sum_{c \in \{90, 70, 40\}} c P_C(c) = 90 \cdot 0.2 + 70 \cdot 0.4 + 40 \cdot 0.4 = 62$$

$$\mathbb{E}[C] = 62$$

(e) What is the standard deviation σ_D of the waiting time?

$$\mathbb{E}[D^2] = \sum_{d=1}^4 d^2 P_D(d) = 1^2 \cdot 0.2 + 2^2 \cdot 0.4 + 3^2 \cdot 0.3 + 4^2 \cdot 0.1 = 6.1$$

$$\mu_d^2 = 5.29$$

$$\sigma_D = 0.9$$

$$\sigma_D \triangleq \sqrt{\mathbb{E}[(D - \mu_D)^2]} = \sqrt{\mathbb{E}[D^2] - (\mathbb{E}[D])^2} = \sqrt{6.1 - 5.29} = \sqrt{0.81} = 0.9$$

Problem 6

6. X is the binomial $(5, 1/2)$ random variable. Find $P_{X|B}(x)$, where the condition $B = \{X > \mu_X\}$. What are $\mathbb{E}[X|B]$ and $\text{Var}[X|B]$?

$$P_X(x) = \begin{cases} \binom{N}{x} p^x (1-p)^{N-x}, & \text{if } x = 0, \dots, N, \\ 0, & \text{otherwise} \end{cases} \quad (N = 5, q = 0.5)$$

$$\mu_X \triangleq \mathbb{E}[X] = Np = 0.5 \cdot 5 = 2.5 \quad B = \{X > 2.5\} = \{3, 4, 5\}$$

$$\Pr[B] = \sum_{x=3}^5 \binom{5}{x} \left(\frac{1}{2}\right)^5 = (10 + 5 + 1) \left(\frac{1}{2}\right)^5 = 16 \left(\frac{1}{2}\right)^5$$

$$P_{X|B}(x) = \begin{cases} \frac{1}{16} \binom{5}{x}, & \text{if } x = 3, 4, 5, \\ 0, & \text{otherwise} \end{cases}$$

Problem 6

6. X is the binomial $(5, 1/2)$ random variable. Find $P_{X|B}(x)$, where the condition $B = \{X > \mu_X\}$. What are $\mathbb{E}[X|B]$ and $\text{Var}[X|B]$?

$$P_{X|B}(x) = \begin{cases} \frac{1}{16} \binom{5}{x}, & \text{if } x = 3, 4, 5, \\ 0, & \text{otherwise} \end{cases}$$

$$\mathbb{E}[X|B] \triangleq \sum_{x \in \mathbb{Z}} x P_{X|B}(x) = \sum_{x=3}^5 \frac{x}{16} \binom{5}{x} = \left(3 \frac{5}{8} + 4 \frac{5}{16} + 5 \frac{1}{16} \right) = \frac{55}{16}$$

$$\mathbb{E}[X^2|B] = \sum_{x \in \mathbb{Z}} x^2 P_{X|B}(x) = \sum_{x=3}^5 \frac{x^2}{16} \binom{5}{x} = \left(9 \frac{5}{8} + 16 \frac{5}{16} + 25 \frac{1}{16} \right) = \frac{195}{16}$$

$$\text{Var}[X|B] \triangleq \sum_{x \in \mathbb{Z}} (x - \mathbb{E}[X|B])^2 P_{X|B}(x) = \mathbb{E}[X^2|B] - (\mathbb{E}[X|B])^2 = \frac{195}{16} - \left(\frac{55}{16} \right)^2 = \frac{95}{256}$$

$$\mathbb{E}[X|B] = \frac{55}{16}, \quad \text{Var}[X|B] = \frac{95}{256}$$

Problem 7

7. X and Y are **independent identical** discrete uniform $(1,10)$ random variables. Let B denote the event that $\max(X, Y) \leq 5$. Find the conditional PMF $P_{X,Y|B}(x, y)$

$$P_X(x) = \begin{cases} 1/10, & \text{if } x = 1, \dots, 10, \\ 0, & \text{otherwise} \end{cases}$$

$$P_Y(y) = \begin{cases} 1/10, & \text{if } y = 1, \dots, 10, \\ 0, & \text{otherwise} \end{cases}$$

Since X and Y are **i.i.d.**, the joint PMF is give by

$$P_{X,Y}(x, y) = P_X(x)P_Y(y) = \begin{cases} 1/100, & \text{if } \begin{matrix} x = 1, \dots, 10 \\ y = 1, \dots, 10 \end{matrix} \\ 0, & \text{otherwise} \end{cases}$$

The probability that $\mathcal{M} \triangleq \max(X, Y) \leq 5$ is

$$\Pr[\mathcal{M}] = \sum_{x=1}^5 \sum_{y=1}^5 P_{X,Y}(x, y) = \frac{25}{100} = \frac{1}{4}$$

Therefore, The conditional PMF $P_{X,Y|B}(x, y)$ is given by

$$P_{X,Y|B}(x, y) \triangleq \Pr[X = x, Y = y | \mathcal{M}] = \frac{\Pr[X = x] \Pr[Y = y]}{\Pr[\mathcal{M}]} = \begin{cases} 1/25, & \text{if } \begin{matrix} x = 1, \dots, 5, \\ y = 1, \dots, 5, \end{matrix} \\ 0, & \text{otherwise} \end{cases}$$

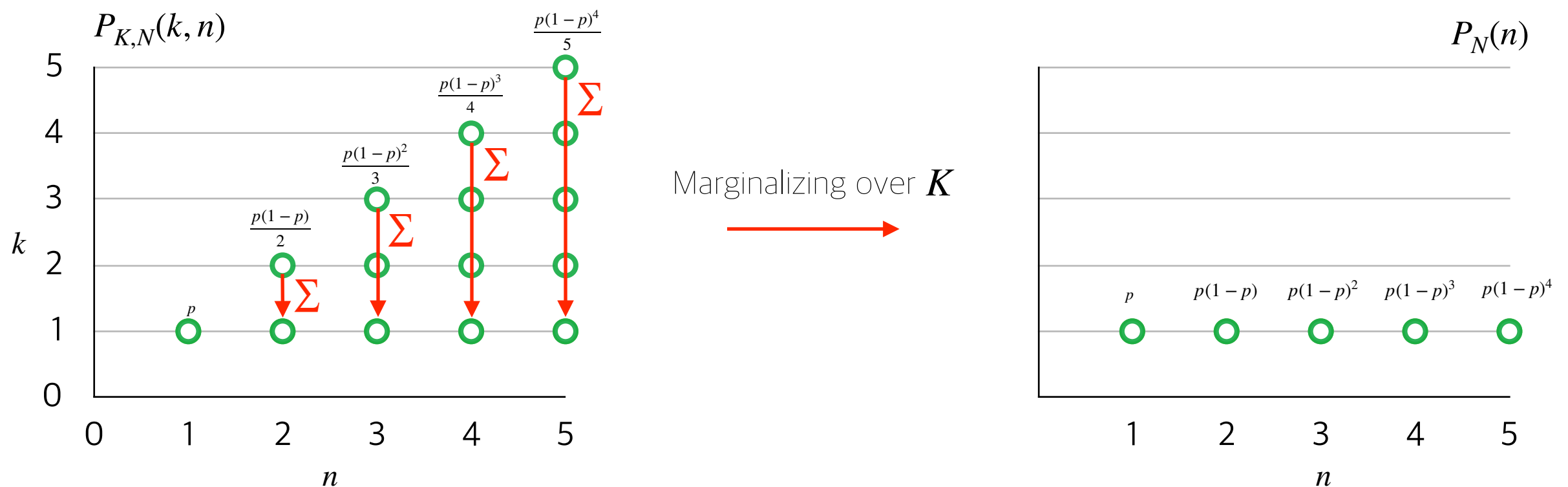
Problem 8

8. N and K have joint PMF

$$P_{K,N}(k, n) = \begin{cases} \frac{p(1-p)^{n-1}}{n}, & \text{if } n = 1, 2, \dots, \\ & k = 1, \dots, n, \\ 0, & \text{otherwise} \end{cases}$$

Let B denote the event that $N \geq 10$.

First, we marginalize over K to obtain the distribution of N



$$P_N(n) = \sum_{k=1}^n P_{K,N}(k, n) = \sum_{k=1}^n \frac{p(1-p)^{n-1}}{n} = \begin{cases} p(1-p)^{n-1}, & \text{if } n = 1, 2, \dots, \\ 0, & \text{otherwise} \end{cases}$$

Problem 8

(a) Find $P_{N|B}(n)$ and $P_{K,N|B}(k, n)$

The probability that B is

$$\Pr[B] = \sum_{n=10}^{\infty} P_N(n) = \sum_{n=10}^{\infty} p(1-p)^{n-1} = p(1-p)^9 \frac{1}{1-(1-p)} = (1-p)^9$$

The conditional PMF $P_{N|B}(n)$ is given by

$$P_{N|B}(n) \triangleq \Pr[N = n | B] = \frac{\Pr[N = n]}{\Pr[B]} = \begin{cases} p(1-p)^{n-10}, & \text{if } n = 10, 11, \dots, \\ 0, & \text{otherwise} \end{cases}$$

The conditional PMF $P_{N,K|B}(n, k)$ is given by

$$P_{K,N|B}(k, n) \triangleq \Pr[K = k, N = n | B] = \frac{\Pr[K = k, N = n]}{\Pr[B]} = \begin{cases} \frac{p(1-p)^{n-10}}{n}, & \text{if } \begin{matrix} n = 10, 11, \dots, \\ k = 1, \dots, n, \end{matrix} \\ 0, & \text{otherwise} \end{cases}$$

$$P_{N|B}(n) = \begin{cases} p(1-p)^{n-10}, & \text{if } n = 10, 11, \dots, \\ 0, & \text{otherwise} \end{cases}, \quad P_{K,N|B}(k, n) = \begin{cases} \frac{p(1-p)^{n-10}}{n}, & \text{if } \begin{matrix} n = 10, 11, \dots, \\ k = 1, \dots, n, \end{matrix} \\ 0, & \text{otherwise} \end{cases}$$

Problem 8

(b) Find $\mathbb{E}[N \mid B]$, $\mathbb{E}[K \mid B]$, $\mathbb{E}[N + K \mid B]$, $\text{Var}[N \mid B]$, $\text{Var}[K \mid B]$

$$P_{N|B}(n) = \begin{cases} p(1-p)^{n-10}, & \text{if } n = 10, 11, \dots, \\ 0, & \text{otherwise} \end{cases}, \quad P_{K,N|B}(k, n) = \begin{cases} \frac{p(1-p)^{n-10}}{n}, & \text{if } n = 10, 11, \dots, \\ 0, & \text{otherwise} \end{cases} \quad k = 1, \dots, n,$$

$$\begin{aligned}\mathbb{E}[N|B] &= \sum_{n=10}^{\infty} nP_{N|B}(n) = \sum_{n=10}^{\infty} np(1-p)^{n-10} \quad \text{Let } z \triangleq n-9 \\ &= \sum_{z=1}^{\infty} (n+9)p(1-p)^{n-1} = 9 \underbrace{\sum_{z=1}^{\infty} p(1-p)^{n-1}}_{=1} + \underbrace{\sum_{z=1}^{\infty} np(1-p)^{n-1}}_{=\frac{1}{p}} \\ &= 9 + \frac{1}{p}\end{aligned}$$

$$\mathbb{E}[N|B] = 9 + \frac{1}{p}$$

Problem 8

(b) Find $\mathbb{E}[N | B]$, $\mathbb{E}[K | B]$, $\mathbb{E}[N + K | B]$, $\text{Var}[N | B]$, $\text{Var}[K | B]$

$$P_{N|B}(n) = \begin{cases} p(1-p)^{n-10}, & \text{if } n = 10, 11, \dots, \\ 0, & \text{otherwise} \end{cases}, \quad P_{K,N|B}(k, n) = \begin{cases} \frac{p(1-p)^{n-10}}{n}, & \text{if } n = 10, 11, \dots, \\ 0, & \text{otherwise} \end{cases} \quad k = 1, \dots, n,$$

$$\mathbb{E}[K | B] = \sum_{n=10}^{\infty} \sum_{k=1}^n k P_{N,K|B}(n, k)$$

$$= \sum_{n=10}^{\infty} \sum_{k=1}^n k \frac{p(1-p)^{n-10}}{n} = \sum_{n=10}^{\infty} \frac{p(1-p)^{n-10}}{n} \sum_{k=1}^n k = \sum_{n=10}^{\infty} \frac{p(1-p)^{n-10}}{n} \frac{n(n+1)}{2}$$

$$= \frac{1}{2} \sum_{n=10}^{\infty} (n+1)p(1-p)^{n-10} = \frac{1}{2} \sum_{n=10}^{\infty} np(1-p)^{n-10} + \frac{1}{2} \sum_{n=10}^{\infty} p(1-p)^{n-10}$$

$$= \mathbb{E}[N | B] = 9 + \frac{1}{p}$$

$$= 1$$

$$= 5 - \frac{1}{2p}$$

$$\mathbb{E}[K | B] = 5 + \frac{1}{2p}$$

Problem 8

(b) Find $\mathbb{E}[N | B]$, $\mathbb{E}[K | B]$, $\mathbb{E}[N + K | B]$, $\text{Var}[N | B]$, $\text{Var}[K | B]$

$$\mathbb{E}[N + K | B] = \mathbb{E}[N | B] + \mathbb{E}[K | B] = 14 + \frac{3}{2p}$$

$$\begin{aligned}\mathbb{E}[N^2 | B] &= \sum_{n=10}^{\infty} n^2 P_{N|B}(n) \\ &= \sum_{n=10}^{\infty} n^2 (1-p)^{n-10} p\end{aligned}$$

$$\begin{aligned}\sum_{m=0}^{\infty} (1-p)^m &= \frac{1}{p}, \\ \sum_{m=0}^{\infty} m(1-p)^m &= \frac{1-p}{p^2}, \\ \sum_{m=0}^{\infty} m^2(1-p)^m &= \frac{(1-p)(2-p)}{p^3}.\end{aligned}$$

Let $m = n - 10$.

$$\begin{aligned}&= p \sum_{m=0}^{\infty} (m+10)^2 (1-p)^m \\ &= p \sum_{m=0}^{\infty} (m^2 + 20m + 100) (1-p)^m \\ &= p \sum_{m=0}^{\infty} m^2 (1-p)^m + 20p \sum_{m=0}^{\infty} m(1-p)^m + 100p \sum_{m=0}^{\infty} (1-p)^m \\ &= \left(\frac{2-3p+p^2}{p^2} + \frac{20-20p}{p} + 100 \right) - \left(81 + \frac{18}{p} + \frac{1}{p^2} \right)\end{aligned}$$

Problem 8

(b) Find $\mathbb{E}[N | B]$, $\mathbb{E}[K | B]$, $\mathbb{E}[N + K | B]$, $\text{Var}[N | B]$, $\text{Var}[K | B]$

$$\begin{aligned}\mathbb{E}[N^2 | B] &= p \cdot \frac{(1-p)(2-p)}{p^3} + 20p \cdot \frac{1-p}{p^2} + 100p \cdot \frac{1}{p} \\ &= \frac{(1-p)(2-p)}{p^2} + \frac{20(1-p)}{p} + 100\end{aligned}$$

$$\begin{aligned}\text{Var}(N | B) &= \mathbb{E}[N^2 | B] - (\mathbb{E}[N | B])^2 \\ &= \left(\frac{(1-p)(2-p)}{p^2} + \frac{20(1-p)}{p} + 100 \right) - \left(9 + \frac{1}{p} \right)^2\end{aligned}$$

$$\text{Var}(N | B) = \frac{1-p}{p^2}$$

Problem 8

- Note:

$$\begin{aligned}\mathbb{E}[N^2] &= \sum_{n=1}^{\infty} n^2 P_N(n) \\ &= \sum_{n=1}^{\infty} n^2 (1-p)^{n-1} p\end{aligned}$$

Let $m = n - 1$.

$$\begin{aligned}\mathbb{E}[N^2] &= p \sum_{m=0}^{\infty} (m+1)^2 (1-p)^m \\ &= p \sum_{m=0}^{\infty} (m^2 + 2m + 1) (1-p)^m \\ &= p \sum_{m=0}^{\infty} m^2 (1-p)^m + 2p \sum_{m=0}^{\infty} m (1-p)^m + p \sum_{m=0}^{\infty} (1-p)^m\end{aligned}$$

$$\begin{aligned}\mathbb{E}[N^2] &= p \cdot \frac{(1-p)(2-p)}{p^3} + 2p \cdot \frac{1-p}{p^2} + p \cdot \frac{1}{p} \\ &= \frac{(1-p)(2-p)}{p^2} + \frac{2(1-p)}{p} + 1 \\ &= \frac{2-3p+p^2}{p^2} + \frac{2-2p}{p} + 1 \\ &= \frac{2-3p+p^2}{p^2} + \frac{2p-2p^2}{p^2} + \frac{p^2}{p^2} \\ &= \frac{2-p}{p^2}\end{aligned}$$

$$\sum_{m=0}^{\infty} (1-p)^m = \frac{1}{p},$$

$$\sum_{m=0}^{\infty} m(1-p)^m = \frac{1-p}{p^2},$$

$$\sum_{m=0}^{\infty} m^2(1-p)^m = \frac{(1-p)(2-p)}{p^3}.$$

$$\text{Var}(N) = \mathbb{E}[N^2] - \mathbb{E}[N] = \frac{1-p}{p^2}$$

$\text{Recall } \text{Var}(N \mid B) = \frac{1-p}{p^2}$

Problem 8

(b) Find $\mathbb{E}[N | B]$, $\mathbb{E}[K | B]$, $\mathbb{E}[N + K | B]$, $\text{Var}[N | B]$, $\text{Var}[K | B]$

$$\begin{aligned}
 \mathbb{E}[K^2 | B] &= \sum_{n=10}^{\infty} \sum_{k=1}^n k^2 P_{N,K|B}(n, k) \\
 &= \sum_{n=10}^{\infty} \sum_{k=1}^n k^2 \cdot \frac{(1-p)^{n-10} p}{n} \\
 &= \sum_{n=10}^{\infty} \frac{(1-p)^{n-10} p}{n} \sum_{k=1}^n k^2 \\
 &= \sum_{n=10}^{\infty} \frac{(1-p)^{n-10} p}{n} \cdot \frac{n(n+1)(2n+1)}{6} \\
 &= \frac{p}{6} \sum_{n=10}^{\infty} (1-p)^{n-10} (n+1)(2n+1)
 \end{aligned}$$

Let $m = n - 10$.

$$\begin{aligned}
 &= \frac{p}{6} \sum_{m=0}^{\infty} (1-p)^m (m+11)(2m+21) \\
 &= \frac{p}{6} \sum_{m=0}^{\infty} (1-p)^m (2m^2 + 43m + 231) \\
 &= \frac{p}{6} \left(2 \sum_{m=0}^{\infty} m^2 (1-p)^m + 43 \sum_{m=0}^{\infty} m (1-p)^m + 231 \sum_{m=0}^{\infty} (1-p)^m \right)
 \end{aligned}$$

$$\begin{aligned}
 \sum_{m=0}^{\infty} (1-p)^m &= \frac{1}{p}, \\
 \sum_{m=0}^{\infty} m (1-p)^m &= \frac{1-p}{p^2}, \\
 \sum_{m=0}^{\infty} m^2 (1-p)^m &= \frac{(1-p)(2-p)}{p^3}
 \end{aligned}$$

Problem 8

(b) Find $\mathbb{E}[N | B]$, $\mathbb{E}[K | B]$, $\mathbb{E}[N + K | B]$, $\text{Var}[N | B]$, $\text{Var}[K | B]$

$$\begin{aligned}\mathbb{E}[K^2 | B] &= \frac{p}{6} \left(2 \cdot \frac{(1-p)(2-p)}{p^3} + 43 \cdot \frac{1-p}{p^2} + 231 \cdot \frac{1}{p} \right) \\ &= \frac{115p^2 + 13p + 2}{3p^2}\end{aligned}$$

$$\begin{aligned}\text{Var}(K | B) &= \mathbb{E}[K^2 | B] - (\mathbb{E}[K | B])^2 \\ &= \frac{115p^2 + 13p + 2}{3p^2} - \left(\frac{10p + 1}{2p} \right)^2 \\ &= \frac{80p^2 + 14p + 5}{12p^2}\end{aligned}$$

$$\text{Var}(K | B) = \frac{80p^2 + 14p + 5}{12p^2} = \frac{5}{12p^2} + \frac{7}{6p} + \frac{20}{3}$$

Problem 9

9. Select integrated circuits, test them in sequence until you find the first failure, and then stop. Let N be the number of tests. All tests are independent, with probability of failure $p = 0.1$. Consider the condition $B = \{N \geq 20\}$

(a) Find the PMF $P_N(n)$

$$P_N(n) = \begin{cases} p(1-p)^{n-1}, & \text{if } n = 1, 2, \dots, \\ 0, & \text{otherwise} \end{cases}$$

(b) Find the PMF $P_{N|B}(n)$

$$P_{N|B}(n) = \Pr(N = n \mid B) = \frac{\Pr(N = n)}{\Pr(B)} = \frac{(0.9)^{n-1}(0.1)}{(0.9)^{19}} = (0.9)^{n-20}(0.1).$$

$$P_{N|B}(n) = \begin{cases} (0.9)^{n-20}(0.1), & n = 20, 21, 22, \dots, \\ 0, & \text{otherwise} \end{cases}$$

Problem 9

9. Select integrated circuits, test them in sequence until you find the first failure, and then stop. Let N be the number of tests. All tests are independent, with probability of failure $p = 0.1$. Consider the condition $B = \{N \geq 20\}$

(c) Find the PMF $\mathbb{E}[N | B]$

$$N | B \stackrel{d}{=} 19 + N', \quad N' \sim \text{Geometric}(0.1),$$

$$\mathbb{E}[N | B] = 19 + \mathbb{E}[N'].$$

$$\mathbb{E}[N'] = \frac{1}{p} = \frac{1}{0.1} = 10,$$

$$\mathbb{E}[N | B] = 29.$$

Problem 10

10. X is the binomial $(5,0.5)$ random variable.

(a) Find the standard deviation of X

$$P_X(x) = \begin{cases} \binom{N}{x} p^x (1-p)^{N-x}, & \text{if } x = 0, \dots, N, \\ 0, & \text{otherwise} \end{cases} \quad (N = 5, p = 0.5)$$

$$\mu_X \triangleq \mathbb{E}[X] = Np = 0.5 \cdot 5 = 2.5$$

$$\mathbb{E}[X^2] = \sum_{x=0}^N x^2 \binom{N}{x} p^x (1-p)^{N-x}$$

$$\text{Var}[X] = \mathbb{E}[X^2] - \mu_X^2$$

$$\sigma_X \triangleq \sqrt{\text{Var}[X]}$$

Problem 10

10. X is the binomial $(5, 0.5)$ random variable.

(a) Find the standard deviation of X

$$\mathbb{E}[X^2] = \sum_{x=0}^N x^2 \binom{N}{x} p^x (1-p)^{N-x} = \sum_{x=1}^N x \frac{N!}{(N-x)!(x-1)!} p^x (1-p)^{N-x}$$

Let $k \triangleq x - 1$

$$\begin{aligned} &= \sum_{k=0}^{N-1} (k+1) \frac{N!}{(N-1-k)!(k)!} p^{k+1} (1-p)^{N-1-k} \\ &= \sum_{k=1}^{N-1} k \frac{N!}{(N-1-k)!(k)!} p^{k+1} (1-p)^{N-1-k} + \sum_{k=0}^{N-1} \frac{N!}{(N-1-k)!(k)!} p^{k+1} (1-p)^{N-1-k} \\ &= \sum_{k=1}^{N-1} k \frac{N!}{(N-1-k)!(k)!} p^{k+1} (1-p)^{N-1-k} + \sum_{k=0}^{N-1} \frac{N!}{(N-1-k)!(k)!} p^{k+1} (1-p)^{N-1-k} \\ &= Np \end{aligned}$$

Problem 10

(a) Find the standard deviation of X

$$\mathbb{E}[X^2] = \sum_{k=1}^{N-1} \frac{N!}{(N-1-k)!(k-1)!} p^{k+1} (1-p)^{N-1-k} + Np$$

Let $z \triangleq k-1$

$$= \sum_{z=0}^{N-2} \frac{N!}{(N-2-z)!(z)!} p^{z+2} (1-p)^{N-2-z} + Np$$

$$= N(N-1)p^2 \sum_{z=0}^{N-2} \frac{(N-2)!}{(N-2-z)!(z)!} p^z (1-p)^{N-2-z} + NP = N^2p^2 - Np^2 + Np$$

$(p + (1-p))^{N-2} = 1$

$$\text{Var}[X] = \mathbb{E}[X^2] - \mu_X^2 = (N^2p^2 - Np^2 + Np) - (N^2p^2) = Np(1-p)$$

$$\sigma_X \triangleq \sqrt{\text{Var}[X]} = \sqrt{Np(1-p)} = \frac{\sqrt{5}}{2}$$

Problem 10

(b) What is $\Pr[\mu_X - \sigma_X \leq X \leq \mu_X + \sigma_X]$

$$P_X(x) = \begin{cases} \binom{N}{x} p^x (1-p)^{N-x}, & \text{if } x = 0, \dots, N, \\ 0, & \text{otherwise} \end{cases} \quad (N = 5, q = 0.5)$$

$$\Pr[\mu_X - \sigma_X \leq X \leq \mu_X + \sigma_X] = \Pr[1.3819660113 \leq X \leq 3.6180339887]$$

$$= \Pr[X \in \{2, 3\}] = \sum_{x=2}^3 \binom{5}{x} \frac{1}{2^5} = \frac{5}{8}$$

$$\bullet \mu_X = 2.5 \quad \sigma_X = \frac{\sqrt{5}}{2}$$

$$\Pr[\mu_X - \sigma_X \leq X \leq \mu_X + \sigma_X] = \frac{5}{8}$$